

Jarrell Wei Ern Tan

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Software Engineer with 2+ years of experience developing enterprise web applications using Angular, TypeScript, C#, and ASP.NET technologies. Specialized in building complex form-driven workflows, API integrations, validation systems, and automated UI testing within regulated business filing environments. Experienced in troubleshooting production issues, implementing complex frontend workflows, and supporting Agile software delivery cycles.

EDUCATION

B.S. Computer Science

08/2021 - 05/2023

Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Languages: C#, TypeScript, Python, C/C++, JavaScript

Frontend: Angular, RxJS, Reactive Forms, HTML5, CSS3, Bootstrap

Backend: ASP.NET Core, RESTful APIs, Entity Framework, C# Services, Node.js

Testing: Playwright, UI Automation Testing

Databases: MySQL, SQL Server

Tools & Platforms: Git, Azure DevOps, Agile/Scrum, CI/CD workflows

WORK EXPERIENCE

Software Developer | AstreaX

08/2023 – Present

- Developed enterprise Angular applications using TypeScript, RxJS, and Reactive Forms to support business registration and amendment workflows in regulated filing systems.
- Developed and maintained ASP.NET backend controllers and service-layer logic supporting business filing workflows, API integrations, and database operations using Entity Framework.
- Implemented complex frontend validation logic, dynamic form behavior, and REST API integrations for entity amendments, statutory agent management, and filing submissions.
- Created and applied Entity Framework migrations to support evolving business requirements and maintain database schema consistency across application updates.
- Investigated and resolved production issues across business-critical workflows, improving application stability and reducing recurring user submission errors.
- Built and maintained automated UI testing frameworks using Playwright, improving regression testing coverage for high-risk filing workflows.
- Optimized complex conditional form interactions and workflow behavior for high-volume filing scenarios.
- Collaborated in Agile development environments to deliver new features, troubleshoot defects, and support production releases through Azure DevOps workflows.

Web Developer Intern | SciQuel

06/2022 – 08/2022

- Implemented responsive frontend UI components from design mockups using JavaScript, HTML, and CSS.
- Integrated frontend registration workflows with backend APIs to support user sign-up functionality.
- Collaborated within a large development team to support mobile web application development.

RELEVANT PROJECTS

Compiler for Simple Language, *Academic Project*

Summer 2022

- Designed and implemented a compiler and parser for a custom programming language using linked-list intermediate representations.
- Built execution logic to interpret generated instructions, manage operand memory locations, and support conditional jump operations.
- Developed parsing functionality capable of translating source programs into executable instruction sequences.